

Eddi Xie

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EDUCATION

University of Toronto

Sept 2026 – Nov 2028 (expected)

- *Master of Engineering (+ Co-op) – Mechanical & Industrial Engineering, Robotics Emphasis*
- Coursework: Perception, State Estimation, Motion Planning, Deep Learning, Nonlinear Control

University of British Columbia

May 2026

- *Bachelor of Applied Science - Mechanical Engineering, Data Science Minor - GPA: 3.59*

SKILLS

Languages: C++, Python, C, C#, TypeScript, MATLAB, SQL, R

Robotics & Perception: OpenCV, YOLOv8, object tracking (KCF/CSRT), state machines, 50 Hz control loops, PID, Betaflight/MSP, LeRobot (SO-101)

Systems & Tooling: FastAPI, React/Vite, CMake, pytest, GitHub Actions CI, Git, Raspberry Pi, Arduino, serial/UART, ESC/BLDC/LiPo power, soldering

CAD & Fabrication: SolidWorks, Fusion 360, NX, FDM Printing, milling, lathe, waterjet, G-Code

TECHNICAL PROJECTS

Independent – Dual-Arm Active-Vision Rig (SO-101, LeRobot)

Sept 2026 – Present

- Printed and assembled two 5-DoF arms with TPU compliant fingers; calibrated and teleoperated through LeRobot.

Flight Control Lead – Autonomous Object Tracking Drone Project – [GitHub](#)

Sept 2025 – Sept 2026

- Built a 50 Hz C++ flight-control loop on a Raspberry Pi: 4-mode state machine (idle/takeoff/track/land) with fail-closed safety gates, commanding Betaflight over MSP; hardware-in-the-loop (HIL) bench testing caught a channel-order bug the unit tests had pinned wrong.
- Integrated YOLOv8n detection and OpenCV KCF/CSRT tracking into the loop through a FastAPI gateway; built the React/TypeScript ground station with live video and telemetry.
- Set up CI (9 C++ test binaries, ~190 pytest, vitest) and shipped the stack as a public repo with ADRs and status docs.

Co-Founder, Project Lead – Coddle – [coddle.app](#)

Aug 2025 – Present

- Shipped a computer-vision inventory scanner that parses fridge and receipt photos into structured ingredient data matched against a 1,502-item catalog; scanning became the dominant input path, adding 98.9% of all beta pantry items.
- Built a real-time AI generation pipeline streaming schema-enforced recipes from the Claude API over server-sent events, with database-driven prompt versioning; delivers 3 recipes per request at ~\$0.017 per request.
- Launched to TestFlight closed beta with ~2,000 automated tests behind an enforced 80% coverage gate; traced beta churn through a 25-event PostHog funnel to a single backend validation bug and shipped the fix with 6 regression tests.

Technical Manager - UBC Capstone: 3D Printing Radioactive Image Phantoms, Qurit

Sept 2024 – Apr 2025

- Modified an FDM 3D printer to extrude UV-curable resin, enabling gradient printing of phantoms that mimic real tissue.
- Led team execution across SolidWorks CAD modeling, technical documentation, and software; adapted Python scripts and G-Code to support the hardware changes and optimize print performance.
- Delivered a printer now used with radioisotopes at BC Cancer Research Centre; nominated for top Mech Eng project.

TECHNICAL WORK EXPERIENCE

Engineering Intern, R&D and Project Lead - MYT Shutters, Zhuhai

May 2025 – Aug 2025

- Designed and built a proof-of-concept motorized shutter (CAD, motor sizing, control electronics, RF remote) to test whether a drive unit could fit inside the existing shutter frame; the MVP passed functional testing and was handed to third-party manufacturers as the basis for production development.
- Ran the R&D solo from concept to sourcing, assembly and tuning; advised on a CNC machinery purchase with suppliers.

Management Intern - Hangar One, Teuge

Jul 2023 – Aug 2023

- Expanded operational scope by integrating maintenance procedures into the Hangar One software system; documented hangar chemicals for safety compliance.